GMP 2

Science

May, 2026

setu.ie

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

GMP 2 (A09766)

Short Title:	GMP 2		
Faculty	Science and Computing		
Department	Science		
Cluster	Quality, Regulatory and Compliance		
Author	BWHELAN		
Credits	10	**Level:**	Introductory

Description of Module / Aims

The aim of this module is to develop students competency in identifying, interpreting and applying Quality Control guidelines according to the healthcare/ pharmaceutical regulations. The student will be able to discriminate between defective and conforming ingredients, components and products.

Programmes

`GMPR-0002`	Higher Certificate in Science in Good Manufacturing Practice and Technology (`WD_SGMPT_C`)	2 / 4 / M

Indicative Content

- Introduction to the routes for authorising and marketing medicinal products; variations, International Conference on Harmonisation (ICH)
- Role of Quality Control (QC) unit; Good Laboratory Practice (GLP), USP, BP, EP and JP monographs; the need for robust investigations into out-of-specification results
- Laboratory Information Management Systems (LIMS); reporting technical problems, documenting rejected products
- Sampling: OC curves, sampling plans, sampling tables; review of pharmacopeial test methods; raw material, component and final product inspection
- Product recalls, the role of the Quality, Technical Services and Customer Service departments
- The Product Life Cycle; ICH Q8 Pharmaceutical Development, Q9 Quality Risk Management, Q10 Pharmaceutical Quality System

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Identify the agencies which regulate pharmaceutical and healthcare manufacturing.
2. Interpret a QC test procedure for a component and non-conformance reports.
3. Demonstrate the use of sampling plans and control charts.
4. Plan for a product recall.
5. Apply the approach of the pharmaceutical sector to develop and manufacture medicinal products.

Learning and Teaching Methods

- Lectures
- Case studies

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Case Studies	50%	1,2,5
Assignment	50%	3,4

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/ skill and understanding to carry out practical/professional tasks. Failure to meet the objectives of projects.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/ professional tasks.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Independent Learning		246

Essential Material

- Ammar, G. _Analysis of Reliability and Quality Control_. New York: Wiley, 2013.
- Miller, J.M. and J.B. Crowther. _Analytical Chemistry in a GMP Environment, a practical guide_. New York: John Wiley & Sons, 2000.

Supplementary Material

- "International Conference on Harmonisation." http://www.ich.org/home.html
- Mullins, E. _Statistics for the Quality Control Laboratory_. Cambridge: Royal Society of Chemistry, 2000.